# The sharp weak-star fixed-point stability constant

Literature answer to Remark 3.5 of arXiv:1604.07587

Status-identification packet — agent_lane_12

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**Status.** `literature_already_answered` (full answer).

## Original question

Casini, Miglierina, Piasecki, and Popescu prove in Theorem 3.4 of arXiv:1604.07587 [1] that if $X$ is a predual of $\ell_1$ and

$$(\operatorname{ext} B_{\ell_1})' \subset r B_{\ell_1}, \qquad 0 \leq r < 1,$$

then weak-star fixed-point stability persists under Banach–Mazur perturbations of size less than $2/(1+r)$. Remark 3.5 on source PDF page 8 states that this estimate is sharp for $r = 0$ and leaves the following issue:

> Prove sharpness of the estimate $2/(1+r)$ when $0 < r < 1$.

Equivalently, writing

$$r^*(X) = \inf\{r > 0 : (\operatorname{ext} B_{\ell_1})' \subset r B_{\ell_1}\}$$

and denoting the stability constant by $\gamma^*(X)$, the source supplies

$$\gamma^*(X) \geq \frac{2}{1 + r^*(X)}$$

and asks whether equality holds in the intermediate regime.

## Decisive later answer

The same authors' later paper, *Stability constants of the weak-star fixed point property for the space* $\ell_1$, arXiv:1611.02133 and J. Math. Anal. Appl. 452 (2017), 673–684 [2], explicitly identifies arXiv:1604.07587 as reference [7]. Its introduction restates the lower bound from that paper and says that the main result proves the reverse inequality for $r^*(X) \in (0, 1)$.

Proposition 2.4 constructs, for arbitrarily small error, an equivalent dual renorming that fails the weak-star fixed point property at distortion tending to $2/(1 + r^*(X))$. Theorem 2.5 on supporting PDF page 7 then states:

$$\boxed{\gamma^*(X) = \frac{2}{1 + r^*(X)}}$$

for every predual $X$ of $\ell_1$ having the $\sigma(\ell_1, X)$ fixed-point property. This is exactly the missing sharpness assertion, and in fact gives the exact stability constant for every such predual.

## Scope and provenance

The original open regime $0 < r < 1$ is fully resolved. This is an explicit literature answer rather than an inferred implication: the supporting paper has the same four authors, cites the source, restates its one-sided estimate, and announces and proves the converse. The identification was found by bounded searches using the exact source sentence, constant $2/(1+r)$, paper title, authors, and weak-star stability terminology.

**Human-review recommendation.** Accept as a full literature answer. Check source Remark 3.5 (PDF page 8) and supporting Proposition 2.4 plus Theorem 2.5 (PDF page 7).

## References


[1] E. Casini, E. Miglierina, Ł. Piasecki, and R. Popescu, *Weak-star Fixed Point Property in* $\ell_1$ *and Polyhedrality in Lindenstrauss Spaces*, arXiv:1604.07587 (2016), Remark 3.5.

[2] E. Casini, E. Miglierina, Ł. Piasecki, and R. Popescu, *Stability constants of the weak-star fixed point property for the space* $\ell_1$, arXiv:1611.02133; J. Math. Anal. Appl. **452** (2017), 673–684, Theorem 2.5.